Reishi Mitani

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Summary

Software and systems engineer focused on applied ML systems, data infrastructure, embedded security, and production reliability.
Experienced in integrating machine-learning pipelines into large-scale services, improving observability for high-traffic APIs, and building secure infrastructure for constrained IoT environments.
Currently working at Indeed on taxonomy, classification, and metadata platforms supporting ML-driven features at scale.
Fluent in English and Japanese; advanced Chinese proficiency (HSK6).

Experience

Software Engineer — Indeed Technologies Japan

Apr 2023 – Present

Work on backend platforms and data infrastructure powering ML-driven metadata and taxonomy services.

ML and taxonomy platform integration

- Architected and scaled AI/LLM-powered job and skills taxonomy workflows using occupation-specific questionnaires to capture structured qualification data
- Led a project extracting important attributes such as administrative divisions using transformer models (MMBERT) and AWS SageMaker
- Implemented backend logic to reconcile model outputs with legacy pipelines and ensure safe rollout
- Designed rule-based override and eligibility systems that enabled non-engineering teams to correct misclassified content and improve downstream retrieval signals

Business impact and experimentation

- Drove A/B-tested improvements including an 18% increase in applications started per user, an 11% increase in activated job seekers, and a 9% reduction in low/zero-result search pages
- Partnered with Product, Data Science, and Taxonomy teams to ship cross-market improvements in user and job understanding without performance regressions

Observability & reliability at scale

- Improved logging and monitoring for metadata APIs handling ~800M requests/day
- Led reductions in redundant error logging, lowering Datadog costs and improving signal-to-noise for incidents
- Designed monitoring thresholds and contributed to SLO/latency analysis workflows for on-call engineers
- Optimized taxonomy data pipelines and search/indexing infrastructure, reducing critical sync time by 75% from ~6 minutes to ~1.5 minutes while scaling to thousands of attributes

Data platform modernization

- Led design and migration of taxonomy snapshot artifacts to a structured JSON format
- Reduced artifact size from ~550MB to ~45MB, improving distribution, parsing speed, and debugging workflows
- Coordinated migrations across downstream services and external consumers

Graph database migration

- Led POC and core importer for migrating taxonomy graph data from Neo4j to NebulaGraph
- Implemented large-scale node/relationship import pipelines (~600k nodes, ~4M edges)
- Contributed to schema mapping decisions and validation workflows to ensure data parity during dual-write rollout

Tech stack: Java, Kotlin, Go, MongoDB, Neo4j, AWS, SageMaker, gRPC, Datadog, PyTorch, TensorFlow, Python

Senior Systems Engineer (Freelance) — Scurid

Jul 2022 – Present

Lead embedded systems and backend engineering for Scurid's IoT security and identity platform.

- Primary developer and maintainer of microscurid-c, Scurid's C-based security library for resource-constrained microcontrollers
- Implemented TLS connectivity, cryptographic signature generation and verification, and nanopb protobuf messaging for constrained devices

- Led porting of MicroScurid-C to Sony Spresense (ARM Cortex-M4F), including MQTT signature verification endpoints, cryptographic integration, and hardware demo development
- Designed automated server certificate provisioning via GCP Certificate Authority Service and runtime TLS certificate reloads for zero-downtime rotation in Kubernetes deployments
- Managed GKE deployments, Helm charts, Bitbucket CI/CD pipelines, narrow rollout validation, and production debugging
- Contributed to SSH tunneling, agent log upload systems, gRPC services, Azure IoT Hub integration, InfluxDB/CockroachDB configuration, and DID document support

Tech: C, Go, Embedded Systems, TLS/mTLS, mbedTLS, secp256k1, Protobuf/nanopb, MQTT, Kubernetes/GKE, Helm, GCP, Azure IoT Hub, DID, IoT Security

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Education

M.S. Interdisciplinary Information Studies

University of Tokyo, Apr 2021 – Mar 2023

B.Eng. Systems Innovation

University of Tokyo, Apr 2016 – Mar 2021

Skills

Programming Languages: Java, Kotlin, Go, TypeScript, C, Python, SQL
Infra/Data: MongoDB, Neo4j, InfluxDB, CockroachDB, AWS, GCP, Kubernetes/GKE, Docker, Helm, distributed data pipelines, search/indexing systems
Protocols/Security: gRPC, Protobuf/nanopb, MQTT, LoRaWAN, SSH tunneling, TLS/mTLS, mbedTLS, secp256k1, DID
Specialties: ML integration, AI/LLM taxonomies, classification systems, backend platforms, observability, embedded systems, system migrations, data modeling

Spoken Languages:

English (fluent, TOEFL iBT 117)
Japanese (native)
Chinese (advanced, HSK6)